Keshab Kamal

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Summary

Experienced Developer with expertise in building APIs, chatbots (LangChain and LangGraph) and AI/ML applications. Proficient in Python, FastAPI, React, PyTorch, and Docker. Seeking growth opportunities to apply skills in developing innovative and impactful software solutions.

Education

Kantipur Engineering College 2021-2025
Bachelor in Computer Engineering *Dhapakhel, Lalitpur*

- **Relevant Coursework:** Data Structures and Algorithms (C++), SQL, Software Engineering, AWS

Namuna College of Science and Management 2018-2021
+2 Science *Gaindakot, Nawalparasi*

Projects

AI RAG Chatbot | *Python, FastAPI, Streamlit, LangChain, LangGraph, Pinecone, Groq*

- Developed an AI RAG Chatbot enabling users to query PDF documents through an intelligent chat interface built with Streamlit and FastAPI.
- Implemented a multi-agent orchestration system using LangGraph to manage workflows, including automated routing between local document search (Pinecone) and real-time web search (Tavily).
- Engineered a robust backend for document processing and indexing, integrating a validation "judge" agent to ensure high-accuracy responses.

Pizza Delivery API | *FastAPI, PostgreSQL, SQLAlchemy, JWT, Docker*

- Built a scalable pizza delivery API using FastAPI and PostgreSQL, implementing secure order tracking and user management.
- Integrated JWT authentication with role-based access control, ensuring secure user login and administrative data protection.
- Containerized the application using Docker, simplifying environment setup and ensuring consistent deployment across platforms.

Driver Drowsiness Detection System | *Python, TensorFlow, OpenCV, Pandas, Streamlit*

- Developed a real-time driver drowsiness detection system using a CNN model to classify open and closed eyes from live video streams.
- Implemented eye detection and preprocessing with OpenCV and integrated the model into an interactive Streamlit web application.
- Optimized system performance by reducing false positives and ensuring smooth real-time execution.

Virtual Try-On: Clothing Image Generation | *PyTorch, Diffusers, TorchVision, Transformers, HuggingFace*

- Developed a diffusion-based virtual try-on model to provide a digital clothing try-on experience.
- Integrated Stable Diffusion XL (SDXL) with IP-Adapter and fine-tuned the model for improved garment realism.
- Implemented data augmentation techniques to enrich the training dataset and enhance model robustness.
- Built an interactive interface using Gradio, enabling users to try on clothes digitally.

Technical Skills

Languages: Python, Javascript, C/C++, HTML, CSS
Web Technologies: FastAPI, React, Node.js, Express.js
AI/ML Technologies: TensorFlow, PyTorch, Scikit-Learn, LangChain, LangGraph, Docker
Concepts: Containerization, Web Hosting, Artificial Intelligence, Machine Learning, Neural Networks, API Development, AWS Cloud
Database Systems: PostgreSQL, MongoDB
Version Control:: Git and Github

Leadership Experience

Organizer at KEC LITE 2024 2024
Kantipur Engineering College *Dhapakhel, Lalitpur*

- Supervised a team of volunteers
- Organized and led interactive workshops

Training and Certifications

MERN Stack Training May – Aug 2024
Broadway Infosys *Dhapakhel, Lalitpur*

- Technologies: MongoDB, Express, React, Node.js (MERN)

AWS Training 2024
AWS Cloud Club, Nepal

- Hands-on training on core AWS cloud services and Deployment workflows
- Worked with EC2,S3, VPC and Load Balancing
- Gained experience with cloud security, monitoring, and cost-optimization